

ChemBuddy

Complete Academic & Technical Operating Manual

Version 3.5.2 Edition

The Comprehensive Academic Operating System for Master of Science (MSc) Chemistry Students, University Faculty & Research Scholars

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Target Audience: MSc Chemistry Students, University Faculty & Researchers

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Supported Platforms: Android (API 26+), Web, Linux, macOS, Windows

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Academic Mission Statement

ChemBuddy by Prajwal A Kambar was engineered specifically to address the demanding academic requirements of postgraduate Chemistry curricula—balancing rigorous laboratory organic and inorganic synthesis, multi-spectral characterization, competitive examination preparation (CSIR-NET, GATE, SET), and strict university attendance compliance.

1. Introduction & Academic Architecture Overview

ChemBuddy is an advanced, offline-first academic operating system designed exclusively for Master of Science (MSc) Chemistry students, researchers, and university faculties. Unlike generic student planners or basic scientific calculator tools, ChemBuddy integrates specialized chemoinformatics algorithms, textbook-grade spectroscopy simulation tools, attendance forecasting mathematics, and chemical reasoning systems into a single, unified, privacy-respecting environment.

Core Engineering Principles

- **100% Offline Core Capabilities:** Timetable intelligence, attendance buffer calculations, 16 physical chemistry calculators, Woodward-Hoffmann pericyclic selection rules, spectroscopy databases, reaction prediction, and 2D vector SVG generation execute completely on-device without requiring an internet connection.
- **Curriculum Grounding:** All chemical formulas, reaction mechanisms, spectroscopic tables, and mathematical derivations strictly adhere to IUPAC nomenclature and standard postgraduate reference texts (Silverstein, Pavia, March, Clayden, Carey-Sundberg, Miessler-Tarr).
- **Data Privacy & Local Sovereignty:** Student profiles, attendance records, study notes, and PDF documents remain stored locally in high-performance encrypted storage boxes. Cloud synchronization is entirely optional.
- **Laboratory Ergonomics:** The interface uses a specialized dark-mode palette (#0A0914 canvas, #7C3AED primary purple, #06B6D4 cyan, #F59E0B gold) engineered to protect dark-adapted eyes in laser spectroscopy rooms, darkrooms, and low-light laboratory settings.

2. Installation, System Requirements & Quick Setup

ChemBuddy is distributed as an optimized Android APK and standalone cross-platform application package.

System Requirements

Parameter	Minimum Requirement	Recommended Specification
Operating System	Android 8.0 (Oreo, API Level 26)	Android 11.0+ (API Level 30+)
RAM	2 GB RAM	4 GB+ RAM (for large PDF caching & 3D rendering)
Storage Space	85 MB free disk space	250 MB free disk space (for offline databases)
Display	720 × 1280 pixels	1080 × 2400 pixels (AMOLED recommended)

Installation Steps

1. **Download Release Package:** Download [ChemBuddy-v3.5.2-by-Prajwal-A-Kambar.apk](#) from the official university repository or GitHub release page.

2. **Enable Package Installation:** On Android, navigate to *Settings* → *Security* → *Install Unknown Apps* and authorize your file manager.
3. **Run Package Installer:** Open the downloaded file and select **Install**.
4. **Grant Essential Permissions:**
 - **Notifications:** Required for the 8:30 AM morning timetable briefing, advance class alerts, and evening cognitive insights.
 - **Exact Alarms (Android 12+):** Ensures schedule notifications trigger with minute-precise accuracy.
 - **Storage / Media:** Required for importing syllabus PDFs, research papers, and exporting attendance or reaction dossiers.
5. **Profile Initialization:** On first launch, enter your Full Name, University/College Name (e.g., Bangalore City University), Academic Semester (Semester 1 through 4), and Specialization Branch (Organic, Inorganic, Physical, or Analytical Chemistry).

3. Personalized Onboarding & Decluttered Academic Dashboard

ChemBuddy v3.5.2 introduces a refined, decluttered header layout that balances clean typography with immediate access to daily academic tasks.

Personalized Identity & Greeting Logic

The system replaces all generic greetings with the student's actual name entered during registration (stored in the local repository as `fullName` or extracted from `registerNumber`, defaulting gracefully to `Prajwal`).

The greeting dynamically adapts to the current hour and attendance standing:

- **00:00 – 04:59 (Late Night Synthesis):** "Burning the midnight burner, [Name] 🔬"
- **05:00 – 11:59 (Morning Energy):** "Good morning, [Name] ☀️"
- **12:00 – 16:59 (Afternoon Laboratory):** "In the flow, [Name] 📖"
- **17:00 – 23:59 (Evening Study):** "Good evening, [Name] 🌃"

Decluttered Layout Architecture

To prevent interface clutter on mobile devices, the top header height has been reduced from ~250px to ~65px:

- **Single-Line Compact Greeting:** Rendered with a clean 17px font weight, constrained to a single line with ellipsis truncation to avoid text wrapping.
- **Persistent Quick Actions:** Notification toggle, global search, and profile status icons are positioned in a streamlined horizontal row.
- **Daily Chem Pulse:** Directly below the greeting, students see their active attendance status badge, next scheduled class countdown, and quick-launch shortcuts to the molecular sketcher, calculators, and spectroscopy tools.

4. Ask ChemBuddy (Intelligent Chemical Reasoning & Computational Tutor)

Ask ChemBuddy is a specialized chemical reasoning tutor designed to guide postgraduate students through complex chemistry concepts, mechanism pathways, spectroscopic interpretation, and computational derivations.

Academic Scope & Capabilities

The reasoning engine is tuned specifically for Master of Science curricula and competitive examinations:

- **Organic Reaction Mechanisms:** Detailed breakdowns of nucleophilic/electrophilic substitutions, additions, eliminations, rearrangement mechanisms (Pinacol, Beckmann, Baeyer-Villiger, Hofmann, Curtius), and organometallic catalytic cycles (Heck, Suzuki, Stille, Grubbs).
- **Spectroscopic Deduction:** Step-by-step interpretation of multi-spectral problems combining ^1H NMR, ^{13}C DEPT, FT-IR, and Mass Spectrometry data to deduce constitutional and stereochemical structures.
- **Physical Chemistry Derivations:** Rigorous step-by-step mathematical derivations of thermodynamic potentials, Maxwell relations, rate laws, quantum mechanical operators (particle in a box, harmonic oscillator), and electrochemistry equations.
- **Inorganic Chemistry & Coordination Complexes:** Crystal Field Theory (CFT), Ligand Field Theory (LFT), Jahn-Teller distortion, Tanabe-Sugano diagrams, magnetic susceptibility, and Wade's rules for cluster compounds.

Structured Presentation Format

Responses generated by Ask ChemBuddy follow a disciplined academic format:

1. **Core Principle / Theorem:** Clear statement of the underlying chemical law or concept.
2. **Mathematical / Mechanistic Formulation:** Step-by-step derivation or electron movement sequence.
3. **Reaction Conditions & Reagents:** Stoichiometry, solvents, temperature requirements, and catalysts.
4. **Stereochemical & Regiochemical Outcomes:** Retention, inversion, Markovnikov/anti-Markovnikov, endo/exo preferences.
5. **Exam Summary / Key Takeaways:** Concise summary points optimized for university answer scripts.

5. Reaction Mechanisms & Step-by-Step Pathway Explorer

The Reaction Mechanisms module provides a comprehensive compendium of organic transformations taught across advanced master's programs.

Comprehensive Reaction Catalog

The catalog encompasses all standard postgraduate reaction classes:

Reaction Category	Representative Examples	Key Mechanistic Focus
Nucleophilic Substitution & Elimination	SN1, SN2, SNi, E1, E2, E1cB	Leaving group ability, solvent polarity, carbocation stability, anti-periplanar geometry.
Electrophilic Aromatic Substitution (EAS)	Nitration, Friedel-Crafts Alkylation/Acylation, Halogenation	Arenium ion (sigma complex) intermediate, directing effects of activating/deactivating substituents.
Named Carbonyl Condensations	Aldol, Claisen, Dieckmann, Knoevenagel, Mannich, Michael Addition	Enolate generation, nucleophilic addition to carbonyl, dehydration driving forces.
Rearrangements & Migrations	Beckmann, Hofmann, Curtius, Lossen, Favorskii, Benzilic Acid	Electron-deficient nitrogen/carbon migration, stereospecificity, migratory aptitude.
Oxidation & Reduction	Swern, PCC, Jones, Dess-Martin, Birch Reduction, Wolff-Kishner, Clemmensen	Chemoselectivity, partial vs full oxidation, radical anion mechanisms.
Transition-Metal Catalysis	Suzuki-Miyaura, Heck, Sonogashira, Stille Coupling	Oxidative addition, transmetalation, migratory insertion, reductive elimination cycles.

Step-by-Step Mechanistic Breakdown

Each reaction entry is structured with full mechanistic transparency:

- **Step Number & Transformation:** Clear identification of what chemical event occurs (e.g., *Step 1: Enolate Formation by Base Deprotonation*).
- **Curved Arrow Electron Movement:** Explicit textual and diagrammatic description of electron pair movement from nucleophile to electrophile.
- **Reactive Intermediates:** Identification of carbocations, carbanions, free radicals, carbenes, or cyclic transition states.
- **Driving Force:** Thermodynamic and kinetic factors driving the reaction forward (e.g., release of aromatic stabilization, formation of gaseous CO₂ or N₂, precipitation of salt).

6. Publication-Grade Reaction Export (PDF Dossier, Excel XLV/CSV & SVG)

ChemBuddy v3.5.2 features an updated export engine ([ExportService](#)) designed to produce publication-grade documentation for laboratory records, seminars, research presentations, and departmental reviews.

Export Formats & Structural Specifications

Format	Output Extension	Structure & Content Included
Academic PDF Dossier	<code>.pdf</code>	• 0.5-inch margins with deep purple (#4C1D95) institutional styling. • Reaction equation banner and dedicated SVG vector scheme diagram. • Comprehensive chemical properties grid (Reactants, Reagents, Conditions, Products, Regioselectivity, Stereochemistry, Driving Force). • Step-by-Step mechanism table (Step #, Transformation Description, Curved Arrow Movement & Intermediates). • Applications, synthetic limitations, and running page footer (Page X of Y).
Excel Spreadsheet (XLV/CSV)	<code>.csv</code>	• Preceded by a UTF-8 Byte Order Mark (<code>\uFEFF</code>) to ensure seamless 1-click opening in Microsoft Excel without character corruption. • Full metadata columns including Reactants, Products, Solvents, Catalysts, and Yield. • Dedicated Step-by-Step Mechanism breakdown column detailing electron movement. • Embedded standalone SVG vector markup code for downstream graphic extraction.
Standalone Vector Graphic	<code>.svg</code>	• Pure scalable vector XML markup with standard atom coordinate geometry. • Crisp rendering at any resolution or zoom level for seminar slides and publications.

In-App Export Workflow

1. **Single Reaction Dossier:** Navigate to any reaction in the Reaction Explorer, tap the **Export / Share** icon in the top AppBar, and select your preferred format (PDF Dossier, Excel Spreadsheet, or SVG Graphic).
2. **Full Compendium Export:** On the main Reaction Mechanism catalog screen, tap the **Download** icon in the AppBar to export the entire library as a comprehensive multi-reaction Excel database or master PDF compendium.

7. ChemDraw Pro & Ketcher Canvas (118-Element Molecular Sketcher)

ChemBuddy includes a chemical structure drawing canvas optimized for mobile touchscreens, tablets, and desktop input devices.

118-Element Periodic Table Selection Palette

Tap the element button or the  icon to open the complete 118-element IUPAC periodic table modal:

- **Search & Filter:** Real-time filtering by element symbol (), English name (), or atomic number ().
- **Standard CPK Coloring:** All elements follow standard CPK color conventions (Carbon: dark grey, Oxygen: red, Nitrogen: blue, Sulfur: yellow, Phosphorus: orange, Chlorine: green, Bromine: dark red, Iodine: purple).
- **Coordination Complex Support:** Full access to transition metals, lanthanides, and actinides for organometallic and inorganic coordination geometry sketches.

Canvas Drawing Tools & Gestures

Tool / Icon	Tool Name	Operation & Functionality
	Bonds	Draws single, double, or triple covalent bonds between atoms with automatic angle snapping (30°, 60°, 120°).
	Stereo Bonds	Wedge (solid) and Dash (hashed) bonds for defining R/S stereocenters and 3D spatial orientation.
	Electron Arrow	Curved arrows for drawing arrow-pushing mechanisms, resonance structures, and pericyclic orbital interactions.
	Marquee Selection	Select, drag, rotate, or delete groups of atoms and bonds collectively.
	2D Clean-Up	Standardizes all bond lengths to 55px and snaps bond angles to idealized textbook geometries.
	Formal Charges	Applies formal charges (+1, -1, +2, -2) and exports  records in MDL Molfiles.
	Reaction Arrow	Draws a horizontal reaction arrow separating reactants from products directly on the canvas.
	Pinch & Pan	Two-finger pinch to zoom (0.3x to 3.0x) and two-finger drag to pan across large schemes. Tap  to reset view.

Functional Group Quick-Palette

Attach standard functional groups to any selected atom in one tap:

-OH (Hydroxyl) · **-NH₂** (Amino) · **-COOH** (Carboxylic Acid) · **-CHO** (Aldehyde) · **-NO₂** (Nitro) · **-CN** (Cyano)

Pre-Built Molecular Templates

Load complex scaffolds without overwriting existing canvas structures: **Benzene, Cyclohexane, Cyclopentane, Pyridine, Aspirin, Naphthalene, Indole, Imidazole, Thiophene, Furan, 18-Crown-6, Ferrocene, Steroid core (cyclopentanoperhydrophenanthrene), Protecting groups (Boc, Fmoc, TBS), and Porphyrin macrocycle.**

8. Reaction Product Prediction & 2D Vector SVG Generator

ChemBuddy v3.5.2 features an offline chemical reaction prediction engine coupled with an on-device vector graphics generator.

Deterministic Offline Reaction Engine

When reactants are drawn on the canvas or provided via SMILES notation and the user taps **Predict Product** ⚡, the system evaluates the transformation against its local rule database:

- **Pharmaceutical Syntheses:** Salicylic Acid + Acetic Anhydride $\rightarrow$ Aspirin; 4-Aminophenol + Acetic Anhydride $\rightarrow$ Paracetamol; Aniline + Acetic Anhydride $\rightarrow$ Acetanilide.
- **Esterification:** Benzoic Acid / Carboxylic acids + Alcohols (with acid catalyst) $\rightarrow$ Esters (Methyl benzoate, Ethyl benzoate, Ethyl acetate).
- **Electrophilic Aromatic Substitution (EAS):** Nitration, Bromination, Sulfonation, Friedel-Crafts Alkylation & Acylation of Benzene, Toluene, Aniline, and Phenol.
- **Named Condensations:** Kolbe-Schmitt, Reimer-Tiemann, Williamson Ether Synthesis, Aldol Condensation, Cannizzaro, Benzoin, Diels-Alder cycloadditions.
- **Functional Group Interconversions:** Carbonyl reductions (NaBH_4 , LiAlH_4), alcohol oxidations (PCC, Jones), and nucleophilic additions.

Client-Side 2D Vector SVG Engine ([SmilesSvgGenerator](#))

To eliminate reliance on external web APIs and avoid pixelated bitmap images, ChemBuddy employs an internal SVG layout engine:

1. **Textbook Coordinate Calibration:** Uses calibrated 2D coordinates for all major ring systems, preserving standard aromatic orientations.
2. **Graph Layout Parser:** Converts arbitrary SMILES strings into connected molecular graphs with 120° zig-zag aliphatic chains and double/triple bond offsets.
3. **Heteroatom Masking:** Draws circular background masks behind heteroatom labels (N, O, S, P, Halogens) to cleanly occlude intersecting bond lines.
4. **Resolution Independence:** Generates clean SVG markup that scales infinitely without degradation on high-DPI screens or printed media.

9. Spectroscopy & Chromatography Suite (8 Analytical Techniques)

The Spectroscopy Hub covers the 8 core analytical and spectroscopic methods central to postgraduate chemical analysis.

The 8 Analytical Techniques

1. **Reverse-Phase HPLC (Chromatography):** Retention factor (k'), resolution (R_s), theoretical plate count (N), and peak asymmetry/tailing (T_f).
2. **Gas Chromatography (Chromatography):** Kovats retention index (I), Van Deemter equation ($H = A + B/u + Cu$), and carrier gas optimization.
3. **Thin Layer Chromatography (Chromatography):** Retention factor ($R_f = d_{\text{solute}} / d_{\text{solvent}}$) and mobile phase polarity screening.
4. **500 MHz 1H NMR (Spectroscopy):** Chemical shift values (δ , ppm), scalar coupling constants (J , Hz), proton integration, and multiplicity splitting patterns.
5. **13C DEPT NMR (Spectroscopy):** Carbon-13 chemical shifts with DEPT-45, DEPT-90, and DEPT-135 discrimination of quaternary, CH, CH₂, and CH₃ carbons.
6. **FT-IR Spectroscopy (Spectroscopy):** Characteristic functional group stretching frequencies ($\bar{\nu}$, cm^{-1}), Hooke's law calculations, and reduced mass (μ).
7. **Mass Spectrometry (Spectroscopy):** Molecular ion (M^+), base peak, isotope abundance clusters ($M+1$, $M+2$ for Cl and Br), and fragmentation pathways (McLafferty rearrangement, alpha-cleavage).
8. **UV-Visible Spectroscopy (Spectroscopy):** Beer-Lambert law ($A = \epsilon c l$), λ_{max} estimation via Woodward-Fieser rules for conjugated dienes and enones.

Interactive Spectrogram Navigation

- **2D Pan & Pinch Zoom (1.0x to 5.0x):** Smooth multi-touch inspection of fine splitting patterns and multiplet peaks.
- **Dedicated Zoom Toolbar:** On-screen controls for Zoom In (+), Zoom Out (-), and View Reset (1.0x).
- **Haptic Peak Snapping:** Tapping near any spectral peak engages haptic feedback and displays a precise crosshair with exact chemical shift (δ), retention time (t_R), or wavenumber (cm^{-1}).
- **Mathematical Formulations:** All governing equations and physical definitions are rendered in standard mathematical typography.

10. Spectroscopy Structure Solver Intelligence (52+ Benchmark Molecules)

The Structure Solver is an algorithmic and database-driven deduction system that assists students in solving unknown molecular structures from multi-spectral data.

Curated MSc Chemistry Database (52+ Compounds)

Contains complete spectral fingerprints (^1H NMR, ^{13}C NMR, FT-IR, MS, UV-Vis) for over 52 postgraduate benchmark molecules: *Aspirin*, *Paracetamol*, *Benzoic Acid*, *Salicylic Acid*, *Benzaldehyde*, *Phenol*, *Aniline*, *Nitrobenzene*, *Toluene*, *Benzyl Alcohol*, *Acetophenone*, *Benzocaine*, *1-Bromopropane*, *2-Bromopropane*, *Vanillin*, *Ethyl propionate*, *Methyl salicylate*, *Ibuprofen*, *Pyridine*, etc.

4-Step Algorithmic Deduction Framework

When an arbitrary molecular formula (e.g., $\text{C}_8\text{H}_8\text{O}$) is entered, the solver executes a 4-step deduction:

1 Degree of Unsaturation / Double Bond Equivalent (DBE / IHD):

$$\text{DBE} = C + 1 - \frac{H}{2} - \frac{X}{2} + \frac{N}{2}$$

A DBE value of 4 or greater immediately indicates the presence of an aromatic ring (C_6H_5 or substituted core).

2 FT-IR Functional Group Classification:

Identifies key diagnostic bands: carbonyls ($1680\text{--}1750\text{ cm}^{-1}$), hydroxyls ($3200\text{--}3600\text{ cm}^{-1}$), amines ($3300\text{--}3500\text{ cm}^{-1}$), and aromatic stretches ($1500\text{--}1600\text{ cm}^{-1}$).

3 ^1H NMR Chemical Shift & Splitting Analysis:

Distinguishes constitutional isomers based on integration and coupling (e.g., Acetophenone's 3H singlet at $\delta 2.6$ vs Benzaldehyde's 1H singlet at $\delta 9.9$).

4 Constitutional Isomer Elimination:

Explains systematically why alternative isomers matching the formula are ruled out based on the spectral evidence.

11. MSc Chemistry Toolkit & 16 Postgrad Calculators

The Chemistry Toolkit consolidates 16 high-precision calculators across 6 dedicated tabs:

Category Tab	Calculator Name	Governing Mathematical Formulation
Solutions	Molar Mass Calculator	$\text{MW} = \sum n_i \cdot M_i$ (Parses complex formulas such as $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$)
	Molarity Calculator	$M = \frac{m}{\text{MW} \times V_{\text{liters}}}$
	Dilution Assistant	$C_1 V_1 = C_2 V_2$
Acid-Base	pH & pOH Calculator	$\text{pH} = -\log[\text{H}^+]$, $\text{pH} + \text{pOH} = 14$
	Henderson-Hasselbalch	$\text{pH} = \text{pK}_a + \log\left(\frac{[\text{A}^-]}{[\text{HA}]}\right)$
	Buffer Formulation	Calculates exact masses of weak acid and conjugate base required for target buffer pH and ionic strength.
Thermo & Kinetics	Gibbs Free Energy	$\Delta G = \Delta H - T\Delta S$ (Determines reaction spontaneity and equilibrium temperature)
	Arrhenius Rate Law	$k = A e^{-E_a / RT}$, $\ln(k_2/k_1) = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$
Spectroscopy	Beer-Lambert Law	$A = \epsilon \cdot b \cdot c$ (Calculates unknown concentration, path length, or molar absorptivity)
	Photon Energy & Wavelength	$E = h\nu = \frac{hc}{\lambda}$ (Interconverts Joules, eV, nanometers, and wavenumbers)
Electrochemistry	Nernst Equation	$E = E^\circ - \frac{RT}{nF} \ln Q = E^\circ - \frac{0.05916}{n} \log_{10} Q$ (at 298.15 K)
	Standard Cell Potential	$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$
Analytical	HPLC Calibration Curve	Linear regression ($y = mx + c$), correlation coefficient (R^2), and unknown quantification.
	Theoretical Plates (N)	$N = 16 \left(\frac{t_R}{W} \right)^2 = 5.54 \left(\frac{t_R}{W_{1/2}} \right)^2$
	Chromatographic Resolution (R_s)	$R_s = \frac{2(t_{R2} - t_{R1})}{W_1 + W_2}$
	Kovats Retention Index (I)	$I = 100 \left[n + (N-n) \frac{\log t'_{R(x)} - \log t'_{R(n)}}{\log t'_{R(N)} - \log t'_{R(n)}} \right]$

12. Pericyclic Hub & Woodward-Hoffmann FMO Symmetry Engine

Pericyclic reactions are governed by orbital symmetry conservation principles. ChemBuddy incorporates a Frontier Molecular Orbital (FMO) selection engine:

Woodward-Hoffmann Selection Rules Matrix

Reaction Type	Electron System	Thermal (Δ) Mode	Photochemical ($h\nu$) Mode
Electrocyclic Reactions	$4n$ electrons (e.g., 1,3-Butadiene)	Conrotatory (HOMO ψ_2 , C_2 symmetry)	Disrotatory (HOMO ψ_3 , σ symmetry)
	$4n+2$ electrons (e.g., 1,3,5-Hexatriene)	Disrotatory (HOMO ψ_3 , σ symmetry)	Conrotatory (HOMO ψ_4 , C_2 symmetry)
Cycloadditions	$4n$ electrons (e.g., $[2+2]$)	Symmetry Forbidden ($\pi 2s + \pi 2s$)	Symmetry Allowed ($\pi 2s + \pi 2s$)
	$4n+2$ electrons (e.g., $[4+2]$ Diels-Alder)	Symmetry Allowed ($\pi 4s + \pi 2s$, Endo preference)	Symmetry Forbidden ($\pi 4s + \pi 2s$)
Sigmatropic Rearrangements	$[1,3]$ Hydrogen Shift	Symmetry Forbidden (Suprafacial)	Symmetry Allowed (Suprafacial)
	$[1,5]$ & $[3,3]$ (Cope, Claisen)	Symmetry Allowed (Suprafacial, Chair-like transition state)	Symmetry Forbidden

13. Smart Timetable & Attendance Intelligence Engine

The attendance module in ChemBuddy is an analytical forecasting engine designed to ensure strict compliance with university regulations.

The 75% Cutoff Mathematics

Most universities enforce a mandatory **75% attendance threshold** for semester examination eligibility. ChemBuddy computes two critical metrics:

1. Classes Needed to Recover 75% (Target Equation):

If current attendance is below 75%, ChemBuddy solves:

$$\text{Classes Needed} = \max\left(0, \lceil 3 \times \text{Absent} - \text{Present} \rceil\right)$$

This reveals the exact number of consecutive classes a student must attend without missing a single lecture to restore eligibility.

2. Safe Classes That Can Be Skipped (Buffer Equation):

If current attendance is safely above 75%, ChemBuddy solves:

$$\text{Safe Bunks Available} = \max\left(0, \left\lfloor \frac{\text{Present} - 3 \times \text{Absent}}{3} \right\rfloor\right)$$

This informs the student how many lectures they can afford to miss for conferences, medical reasons, or emergencies while remaining strictly at or above 75.0%.

Supported Attendance Statuses

Status	Effect on Percentage	Academic Use Case
Present (✓)	Increments both Present and Total Count	Student attended lecture or practical session.
Absent (✗)	Increments Total Count; Present count unchanged	Student missed lecture or was marked absent.
Postponed / Cancelled (⏸)	Excluded entirely from calculation	Faculty was absent, institutional holiday, or lab rescheduled. Does not penalize the student's attendance.
Excused / On Duty (OD)	Treated as Present	Official university duty, symposium, seminar presentation, or authorized medical leave.

14. Notification System & Action Shade (1-Tap Attendance Tracking)

ChemBuddy features an automated background notification engine synchronized to the **Asia/Kolkata (IST)** timezone, operating across 5 dedicated channels.

⚡ 1-Tap Notification Shade Attendance

Students do not need to open the application to record their attendance. When a class reminder appears in the Android notification shade, tapping **Present** ✓ or **Absent** ✗ logs the attendance record directly to local storage, recalculates attendance percentages, and displays a discreet confirmation toast.

The 5 Notification Channels

Channel Identifier	Trigger Schedule	Description & Academic Purpose
Class Reminders <code>chem_buddy_classes</code>	30 minutes prior to class (configurable: 5, 10, 15, 30, 60m)	Advance alert indicating room number, course title, current attendance percentage, and 1-tap Present / Absent action buttons.
Daily Timetable <code>chem_buddy_daily</code>	Every morning at 8:30 AM IST	Comprehensive morning schedule summary: total lecture count, practical timings, room allocations, and overall attendance standing.
Daily Psychology Facts <code>chem_buddy_psychology_facts</code>	Every evening at 8:00 PM IST	Daily evidence-based cognitive insight into learning psychology, memory consolidation, and focus strategies.
Tests & Deadlines <code>chem_buddy_deadlines</code>	24h & 2h prior to deadline	Alerts for upcoming internal tests, laboratory journal submissions, project presentations, and semester examinations.
Study & Flashcards <code>chem_buddy_flashcards</code>	When spaced review is due	Prompts for spaced repetition flashcard review based on the SuperMemo-2 forgetting curve algorithm.

15. Exporting Attendance Records (Institutional PDF & Excel XLV/CSV)

To facilitate submission to university faculties, academic advisors, and department heads, ChemBuddy provides multi-format attendance report generation.

Institutional Attendance Reports

- **Institutional PDF Report:** Formatted with official university headers, student registration number, department name, subject-wise lecture breakdown (Total, Attended, Missed, Percentage), and eligibility status certification.
- **Excel Spreadsheet (XLV/CSV):** Formatted with a UTF-8 Byte Order Mark (BOM) for compatibility with Microsoft Excel, featuring date-stamped audit logs of every attendance event.

16. Smart Flashcards & Spaced Repetition (SuperMemo-2 Active Recall)

ChemBuddy implements the **SuperMemo-2 (SM-2) algorithm** to support long-term memorization of reaction mechanisms, reagents, and spectroscopic data.

- **Active Recall Testing:** Cards present a prompt or starting material first. Students formulate the product or mechanism mentally before flipping the card.
- **Recall Quality Rating (1 to 4):**
 - **Again (1)** : Complete failure of recall; card repeats in current session.
 - **Hard (2)** : Recalled with significant difficulty; review interval expands slowly.
 - **Good (3)** : Correct recall with normal effort; optimal interval expansion.
 - **Easy (4)** : Instant, confident recall; interval expands significantly.
- **Automated Deck Generation:** Converts uploaded notes and mechanism summaries into interactive flashcard sets.

17. PDF Library, Study Hub & Grounded Academic RAG Assistant

The PDF Library allows students to store, organize, and search their textbooks, syllabus documents, and research papers.

Single-Source Grounding Architecture

- **Strictly Grounded Retrieval:** In the PDF Study Hub, the AI assistant is constrained strictly to the extracted text of the uploaded PDF document.
- **Standardized Refusal:** If a query is outside the scope of the provided document, the assistant explicitly states: *"This topic is not covered in the provided document."* to avoid hallucinated information.
- **Extended Token Output:** Supports comprehensive, detailed answers (up to 8,192 tokens) complete with KaTeX mathematical formatting and structural ASCII diagrams.

18. Exam Mode & BCU Model Answer Generator

Calibrated to Bangalore City University (BCU) and Indian university examination patterns:

- **2-Mark Question Mode:** Concise, precise definitions, IUPAC names, and fundamental statements.
- **5-Mark Question Mode:** Step-by-step mechanisms, curved arrow notations, and intermediate structures.
- **10-Mark Question Mode:** Comprehensive essays, retro-synthetic analysis, multi-step synthesis routes, and spectroscopic confirmation.

19. Daily Cognitive & Learning Science Facts (120+ Curated Insights)

Postgraduate study requires sustained mental focus and emotional resilience. ChemBuddy includes a curated library of 120+ evidence-based cognitive psychology insights:

1. Cognitive Biases & Heuristics

Zeigarnik Effect, Halo Effect, Dunning-Kruger Effect, Spotlight Effect, Framing Effect, Anchoring Bias, Loss Aversion, Sunk Cost Fallacy, and Dual-Process Theory.

2. Memory & Learning Psychology

Ebbinghaus Forgetting Curve, Testing Effect, Serial Position Effect (Primacy & Recency), State-Dependent Memory, Sleep Consolidation, and Generation Effect.

3. Neurobiology & Focus

Dopamine Anticipation, Neuroplasticity, Ultradian Rhythms (90-min cycles), Amygdala Regulation, and Sleep Spindles.

4. Habits & Academic Performance

The 20-Second Rule, Implementation Intentions ("If-Then"), Parkinson's Law, Decision Fatigue, and Daily 1% Compounding.

Prime-Weighted Non-Repeating Distribution Formula

To guarantee that every day presents a distinct, non-repeating insight across the academic calendar, ChemBuddy employs a deterministic distribution formula:

$$\text{Index} = \left(\text{Year} \times 365 + \text{DayOfYear} \times 47 + \text{Month} \times 13 \right) \bmod \text{TotalFacts}$$

20. Laboratory Safety & GHS Hazard Library

Essential for laboratory research scholars and postgraduate practical examinations:

- **GHS Pictograms & Hazard Codes:** Explanations of H-statements (Hazard Statements) and P-statements (Precautionary Statements).
- **Chemical Incompatibility Matrix:** Identifies hazardous reagent combinations (e.g., strong oxidizers with organic solvents, water-reactive organometallics).
- **First Aid Protocols:** Immediate emergency response steps for acid burns, alkali exposure, inhalation of toxic vapors, and solvent spills.

21. Offline Security, Local Encrypted Storage & Data Portability

ChemBuddy operates with a strict local-first data architecture:

- **Encrypted Local Storage:** Student profiles, attendance logs, notes, and preferences are stored in encrypted binary format on the user's device.
- **Complete Data Portability:** Students can export their entire profile, timetable, and attendance records as a structured JSON file at any time for backup.
- **Zero Unsolicited Data Transmission:** The application does not transmit student personal data, attendance logs, or sketcher data to external servers.

22. Troubleshooting & Frequently Asked Questions (FAQ)

Q1: Why did the 8:30 AM morning timetable notification not trigger?

Resolution: On Android 12+, confirm that "Alarms & Reminders" permission is granted in Android System Settings. Additionally, set battery optimization for ChemBuddy to "Unrestricted" so that the system's background Doze mode does not defer morning alarms.

Q2: Can coordination complexes and organometallics be sketched in ChemDraw Pro?

Resolution: Yes. Open the 118-element periodic table palette, select the transition metal center (e.g., Fe, Co, Ni, Pt), and use coordination bonds. Pre-configured templates for *Ferrocene* and *Porphyrin* are also available.

Q3: How does ChemBuddy handle attendance for cancelled or rescheduled lectures?

Resolution: Mark the lecture as *Postponed* / *Cancelled* (). ChemBuddy removes that class slot from both the numerator and denominator, ensuring the student's percentage is not unfairly penalized.

Q4: Does reaction prediction or SVG generation require an active internet connection?

Resolution: No. The offline reaction rule engine and the `SmilesSvgGenerator` vector layout engine run entirely on-device with zero latency.

Q5: How can attendance records and reaction mechanisms be presented to university faculty?

Resolution: Use the built-in export engine to generate either an official Academic PDF Dossier (formatted with margins, headers, and mechanisms) or an Excel Spreadsheet (XLV/CSV) with UTF-8 BOM encoding for direct opening in Microsoft Excel.

ChemBuddy v3.5.2 Operating Manual

Engineered for MSc Chemistry Students, University Faculty & Research Scholars worldwide by **Prajwal A Kambar**.